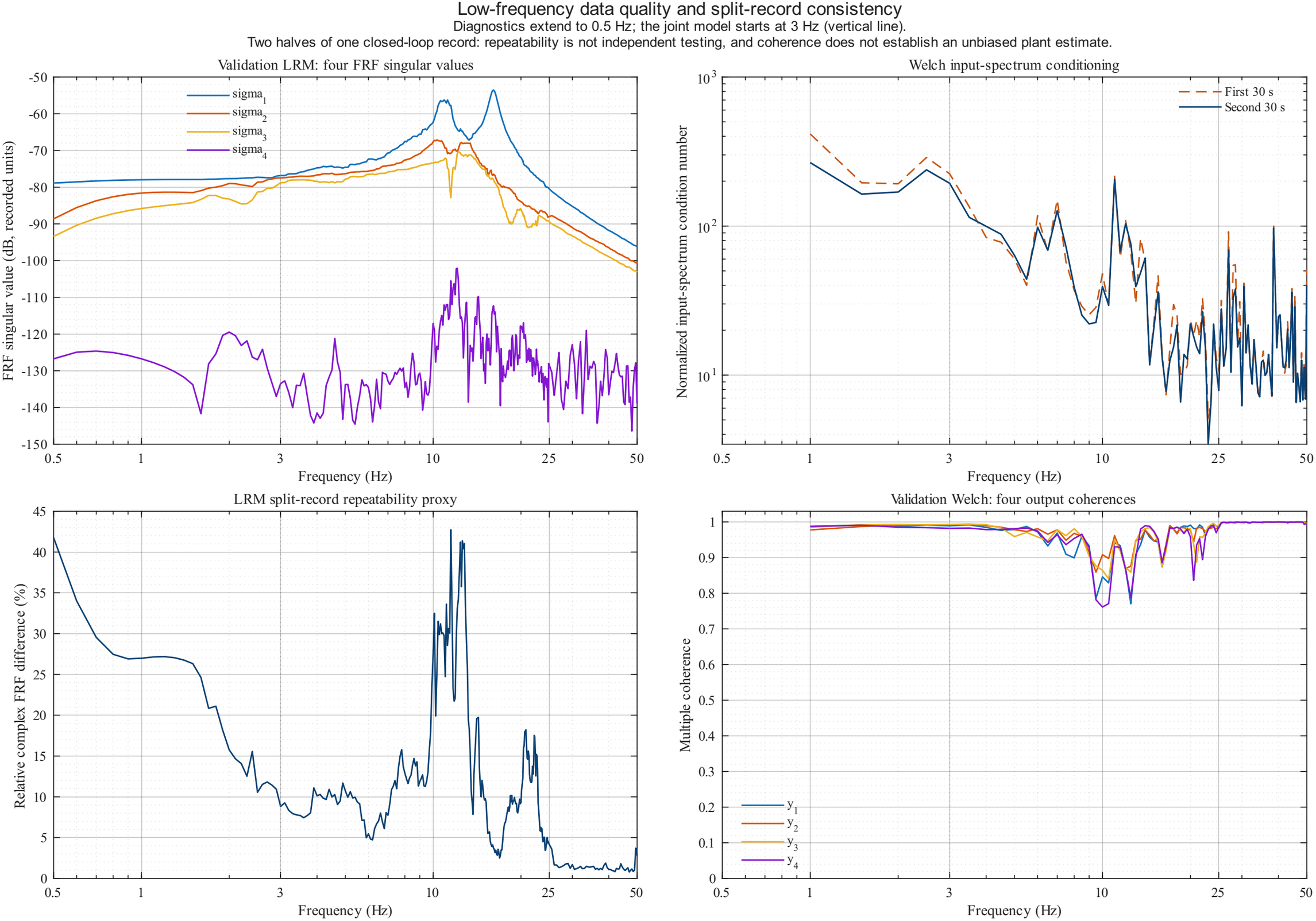

Low-frequency data quality and split-record consistency
Diagnostics extend to 0.5 Hz; the joint model starts at 3 Hz (vertical line).
Two halves of one closed-loop record: repeatability is not independent testing, and coherence does not establish an unbiased plant estimate.
Validation LRM: four FRF singular values
FRF singular value (dB, recorded units)
Frequency (Hz)
sigma1
sigma2
sigma3
sigma4
Welch input-spectrum conditioning
Normalized input-spectrum condition number
First 30 s
Second 30 s
LRM split-record repeatability proxy
Relative complex FRF difference (%)
Validation Welch: four output coherences
Multiple coherence
y1
y2
y3
y4